

PHILOSOPHY OF SCIENCE

LESSON 7

KUHN (AND POPPER) ON SCIENTIFIC PROGRESS - PART 1

Carsten Bergenholtz (cabe@mgmt.au.dk)

Department of Management, Aarhus BSS

Aarhus University

SUMMARY OF LESSON 6

- What is a theory (not)?
- A theoretical explanation
- Different aims of theory: Causality vs. understanding
- Theory vs. models vs. hypotheses
- Theory building and theory testing or different theoretical reasonings: Inductive vs. deductive – and abductive approaches

On theory i) Why we need theories & what are they? ii) Primary aim of theory: Causality or understanding iii) Theory building and testing: Inductive, deductive, and abductive approaches	NYT 2016 Bergenholtz 2026f
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LESSON 6: THEORETICAL EXPLANATION

AN EXAMPLE

TWO DIFFERENT METAPHORS FOR WHAT CONSTITUTES A THEORETICAL EXPLANATION

What does it mean to have 'theoretically explained' something?

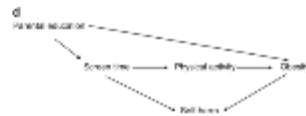
An explanation – the causal logic

- Why did outcome (not) happen? (counterfactual)
- Predicting relationships (probabilistic, on average)
- Causal diagrams can be drawn



An explanation – understanding

- Make sense of a situation
- Subjective meanings/experiences



<https://www.tandfonline.com/doi/full/10.1080/01449133.2017.1345013>

Causality

- Questions: Did remote-work cause higher productivity / more committed employees...?
- Evidence: What confounders to consider?
- Research designs: Establishing counterfactual

Understanding

- Questions: How do remote-work setups shape the work environment and experience? Do workers reinterpret identity, dynamics, relations...?
- Evidence: Workers' stories and how they see the world
- Research designs: Interviews, diaries, observations...

Note: Theoretical explanations and concepts are more abstract than the actual measured phenomena

LESSON 6: DEDUCTIVE VS INDUCTIVE

CLOSED APPROACH

1

Matches the specific bet (6-6-6).

OPEN APPROACH

3

Noteworthy patterns found inductively.

2	3	5	4	5	2	4	3	4	1	3	3	1	3
3	3	6	2	6	4	4	4	6	6	2	2	5	6
5	1	3	6	4	5	2	5	6	5	2	1	1	5
4	5	5	6	6	5	1	2	5	4	6	2	5	1
3	6	4	2	6	3	2	3	6	6	3	1	6	1
6	5	3	5	3	1	6	1	3	5	6	6	6	4
1	3	5	5	4	3	6	3	1	4	2	2	5	1
6	4												

Biased Die
(20%)



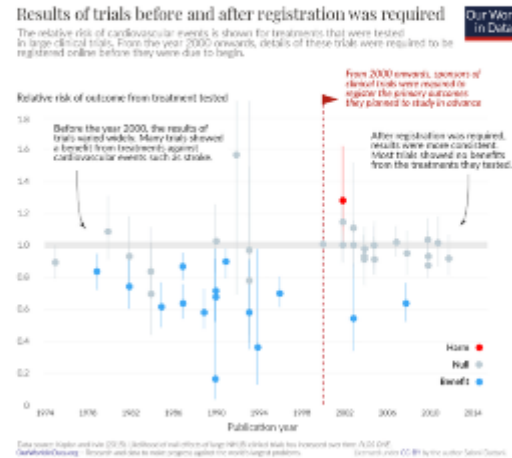
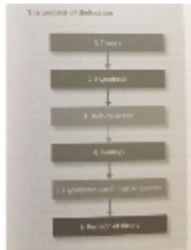
Fair Die
(16.7%)

https://carstenbergenholtz.github.io/Phil-of-Science-tools/open_closed_approach.html

LESSON 6: DEDUCTIVE VS INDUCTIVE

PRE-REGISTERING HYPOTHESES (NEW TREND)

- How does audience know if a hypothesis was derived from theory, or changed after looking at data?
- We do not – unless researchers have pre-registered hypothesis = registered their hypothesis on a closed online platform
- Pre-registration = actual deductive science



Why is a deductive test stricter / "better" than an inductive test?

- Measurement
 - Subjective rating by patient (after 3, 6, 12 weeks e.g.)
 - Physiological measurement by researcher (after 3, 6, 12 weeks e.g.)
- If not pre-registered deductive test, researcher – COULD – pick any outcome that is good (and ignore/downplay others)

LESSON 7: AGENDA

From improving research designs and insights on how to develop theory → more general questions about truth of theories and how science develops

Kuhn and Popper on Scientific Progress - Part 1 i) When is a theory 'true'? Understanding the nature of science and scientific methods ii) How does science develop: Comparing Kuhn's idea of revolutions and Popper's falsification	Kuhn 1970 (section 4 p. 191-198 & section 6 p. 205-207) PoS 5 ABH 2018 ch 3 Popper 1962 (text segments)
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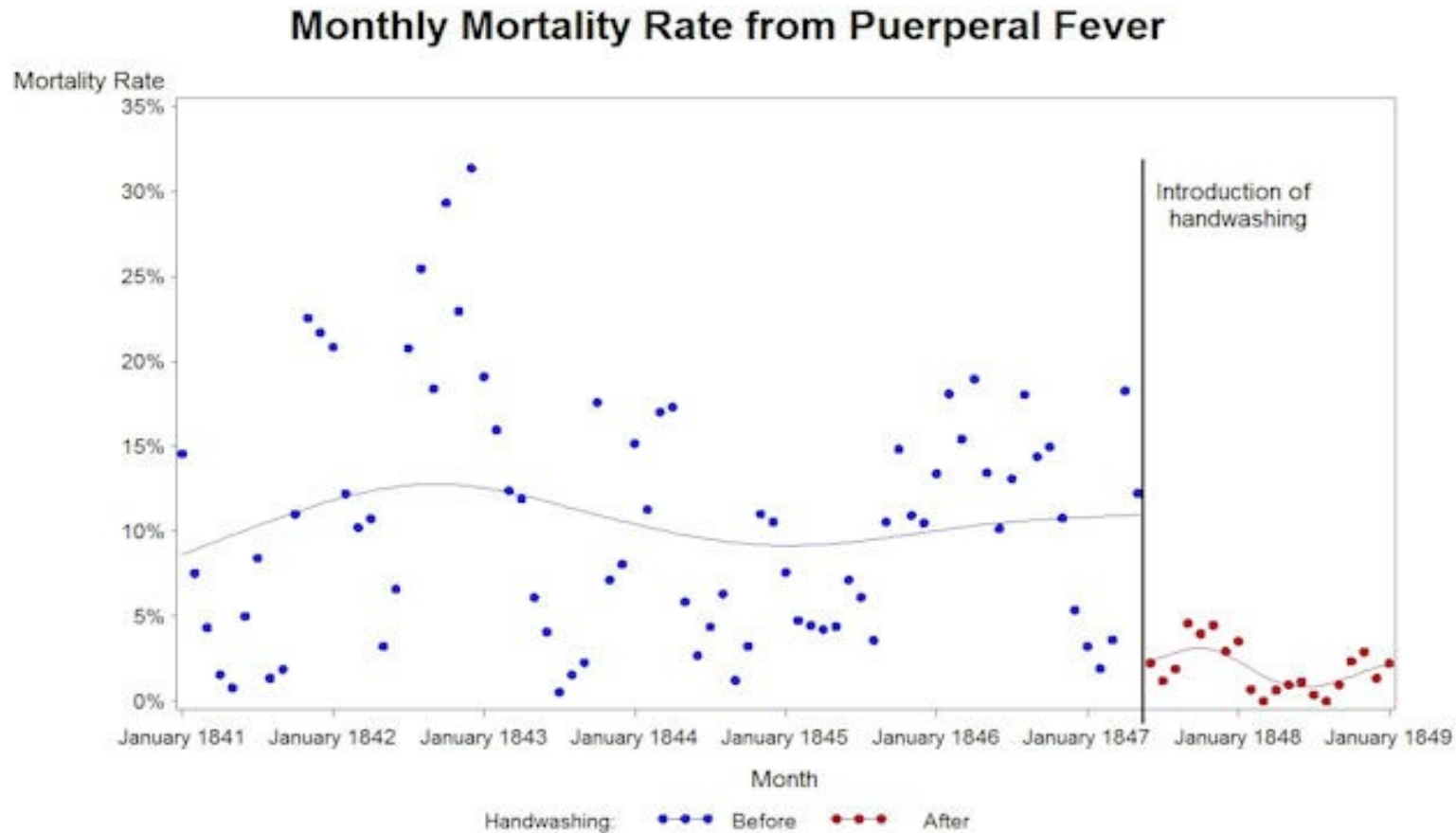
KUHN 1962/70: MOST CITED SOCIAL SCIENCE RESEARCH

(= MOST OFTEN IN REFERENCE LIST)

Book	Author	Date*	Discipline	Citations
The Structure of Scientific Revolutions	Thomas Kuhn	1962	Philosophy	81,311
Diffusion of Innovations	Everett Rogers	1962	Sociology	72,780
Pedagogy of the Oppressed	Paulo Freire	1968/1970	Education	72,359
Competitive Strategy	Michael E Porter	1980	Economics	65,406
Imagined Communities	Benedict Anderson	1983	Political Science	64,167
Mind in Society	LS Vygotsky	1978	Psychology	63,809
Discipline and Punish	Michel Foucault	1976/1977	Philosophy	60,700
A Theory of Justice	John Rawls	1971	Political Science	58,594
Social Foundations of Thought and Action	Albert Bandura	1986	Psychology	55,324
The Interpretation of Cultures	Clifford Geertz	1973	Anthropology	48,984
The History of Sexuality (3 Volumes)	Michel Foucault	1978-1986	Philosophy	47,955
Situated Learning: Legitimate Peripheral Participation	Jean Lave and Etienne Wenger	1991	Education	47,627
The Fifth Discipline	Peter M Senge	1992	Management	43,876
Institutions, Institutional Change and Economic Performance	Douglass North	1990	Economics	43,411
Culture's Consequences	Geert Hofstede	1980	Management	42,144
The Presentation of the Self in Everyday Life	Erving Goffman	1959	Sociology	40,573
Das Kapital	Karl Marx	1867-1894	Economics	40,237
Distinction: A Social Critique of the Judgement of Taste	Pierre Bourdieu	1984	Sociology	39,729
The Social Construction of Reality	Peter Berger and Thomas Luckmann	1966	Sociology	38,845
Metaphors We Live By	George Lakoff and Mark Johnson	1980	Linguistics	38,723
Stress, Appraisal and Coping	Richard Lazarus and Susan Folkman	1984	Psychology	38,665
Communities of Practice	Etienne Wenger	1999	Psychology	37,775
The Economic Institutions of Capitalism	Oliver Williamson	1985	Economics	37,651
Motivation and Personality	Abraham Maslow	1954	Psychology	37,614
Attachment	John Bowlby	1969	Psychology	37,318

HOW RATIONAL IS SCIENCE (DEVELOPMENT)?

SOMETIMES A CLEARLY BETTER EXPLANATION IS NOT ACCEPTED BY RESEARCHERS



Kuhn's core question: If evidence alone doesn't settle big questions quickly, what *actually* explains scientific agreement and scientific change?

Note: Semmelweis had no (good) theory

<http://www.jameslindlibrary.org/articles/how-dramatic-were-the-effects-of-handwashing-on-maternal-mortality-observed-by-ignaz-semmelweis/>

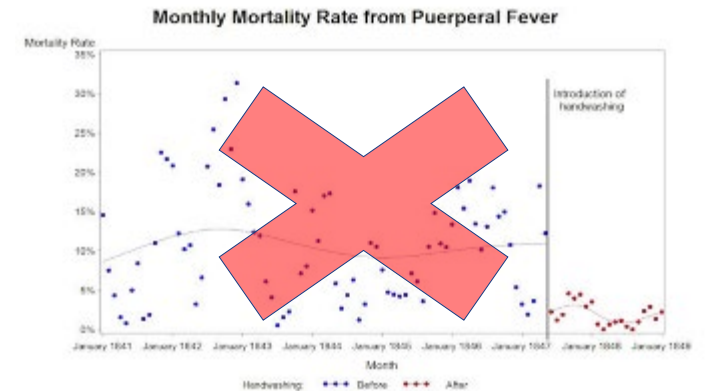
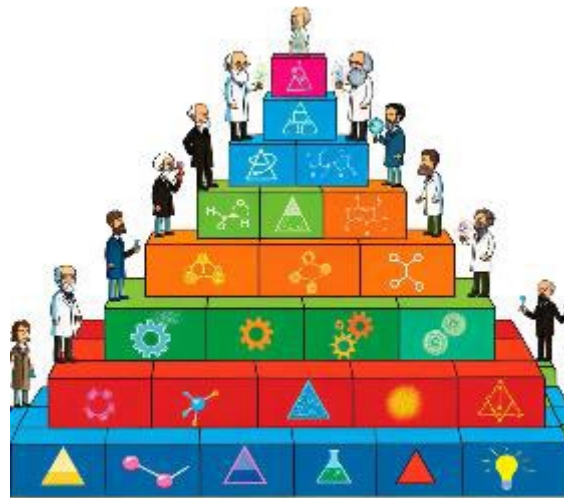
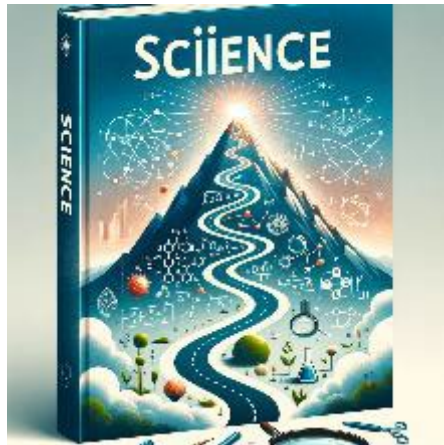
PHILOSOPHY OF SCIENCE: HOW DO THEORIES DEVELOP AND IMPROVE OVER TIME?

Insights on research design and deduction & induction can give impression of:

- Researchers measure world, test theories & change theory if wrong
- Science builds directly on prior work & continuously improves

→ Positivism

Science as a very rational, cumulative process
Evidence settles disagreements... (more in lesson 9)



THE (POSITIVISM) POSITION THAT KUHN & POPPER ARGUE AGAINST



Positivism: Classic 'textbook' conception of science pre-Kuhn (1970)

- “...positivists believed that disputes between rival scientific theories **could be solved in a perfectly objective way**, comparing the theories directly to the observations...” (PoS 5, p.80)
 - The **world can be observed as it is** / there are neutral facts (PoS p. 80)
 - **Theory corresponds to / mirrors** world (PoS p. 78) →
 - We can tell if a theory is true / false
 - Theory development as a **rational process** (PoS p. 78-79)

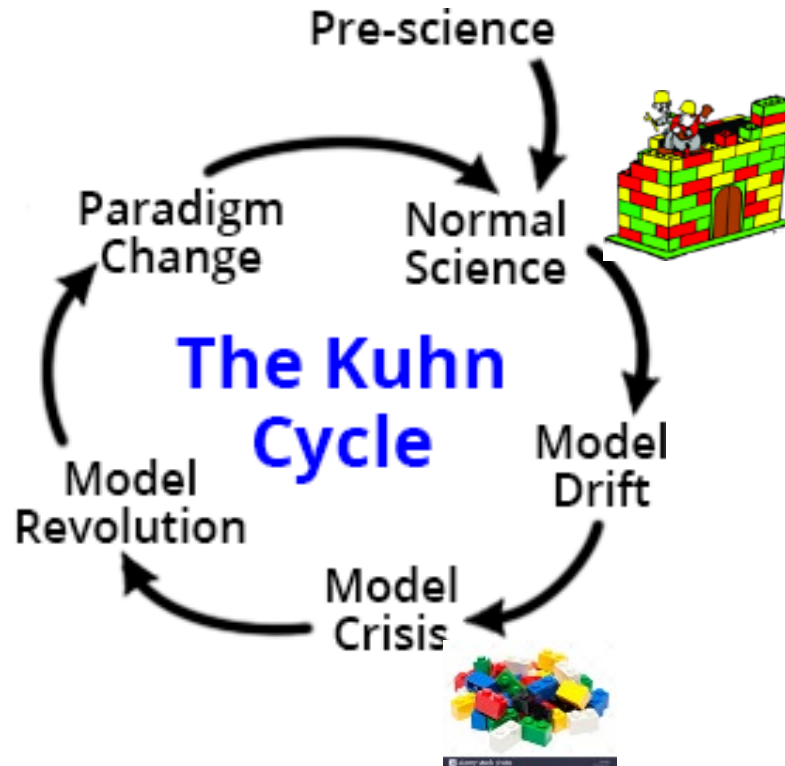
→ Linear Lego brick metaphor: Scientific development is rational and cumulative

Implication: If science is fully rational and theory can be directly compared to observation, then clearly better theories should be *straightforwardly* (even immediately) accepted.

The following introduction of Kuhn is also available in this video from 2021: <https://youtu.be/gD84WMLmQes>

KUHN: 2 TYPES OF SCIENTIFIC DEVELOPMENT I

NORMAL SCIENCE VS. MODEL CRISIS/REVOLUTION



- Core question: How to explain science development and disagreements?
- Most science development is '**normal science**' = lego brick building = fairly **rational** (as in positivism)
- **Kuhn's focus & contribution:** Yet, when new evidence challenges accepted wisdom = **model crisis**, not entirely rational (unlike positivism)

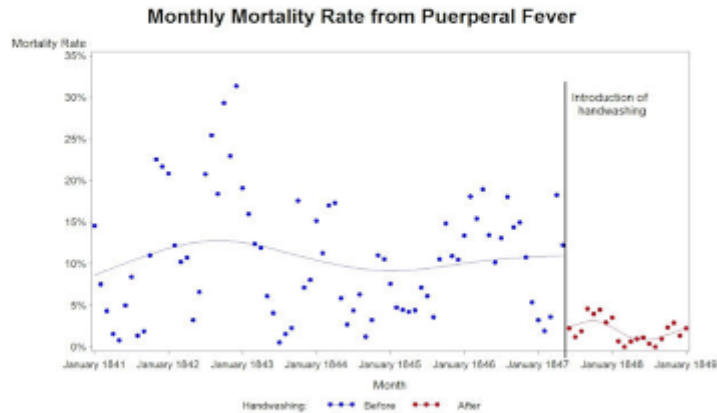
See also title of Kuhn's book: Structure of Scientific Revolutions

<http://www.thwink.org/sustain/glossary/KuhnCycle.htm>

KUHN: 2 TYPES OF SCIENTIFIC DEVELOPMENT II

NORMAL SCIENCE VS. MODEL CRISIS/REVOLUTION

HOW RATIONAL IS SCIENCE (DEVELOPMENT)?
SOMETIMES A CLEARLY BETTER EXPLANATION IS NOT ACCEPTED BY RESEARCHERS



<http://www.jstor.org/stable/2341000>
maternal mortality observed by Ignaz Semmelweis?

Kuhn's core question: If evidence alone doesn't settle big questions quickly, what *actually* explains scientific agreement and scientific change?

Note: Semmelweis had no (good) theory



© Can Stock Photo - csp14722005

Different kinds of developments

- Small: Wash at 30 rather than 28 degrees (**normal science**)
- Big: Wash hands, it prevents death (**model crisis, revolutionary change?**)

A SCIENTIFIC PARADIGM (DEFINITION)

WHAT PARADIGMS ARE AND WHY THEY ARE IMPORTANT

Paradigm: "...a paradigm is an entire scientific outlook – a constellation of shared assumptions, beliefs, and values that unite a scientific community" (PoS p. 81)

Paradigms refer to the way "things are done" (**norms**) & are **implicit** in scientific community

For example: How we usually measure X variable, what questions we ask, what science gets funded, who a science institution would hire, what methods are popular in a field, what qualifies as a good answer etc.

"A paradigm governs, in the first instance, not a subject matter but rather a group of practitioners"
(Kuhn, p. 180)

Paradigms are about fundamental assumptions, thus difficult to change
New evidence – even good evidence – might not change minds (more in lesson 10)

EXAMPLE - IF STUDYING EMOTIONS: HOW PARADIGMS SHAPE HOW SCIENTISTIS PROCESS EVIDENCE

If studying emotions

- If similar approaches (methods, data etc.) fairly easy to compare quality
- Yet, approaches can be very different, making comparison difficult
 - Observing (subjectively) real world situations vs. what happens inside the brain
 - How to decide what is correct / 'objective'?

→ incomparable across different 'paradigms' – different standards

Early Kuhn: Can't compare across paradigms

Later Kuhn: Yeah ok, you can...somehow (more later)

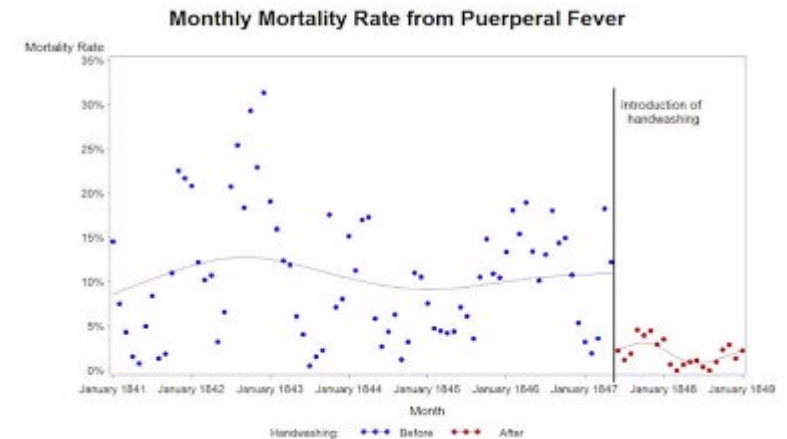


OVERALL: PARADIGMS SHAPE WHAT COUNTS AS GOOD EVIDENCE (PERCEPTION AND NORMS)

Based on a review of **the history of science**, Kuhn presents two explanations for why researchers **resist** new scientific evidence / theories

- **Psychology:** Human perception/cognition shape what evidence we see
- **Sociology:** Norms shape what is considered evidence (no neutral observation or assessment of evidence)

In other words: How researchers are stubborn and are not always convinced by evidence presented b others

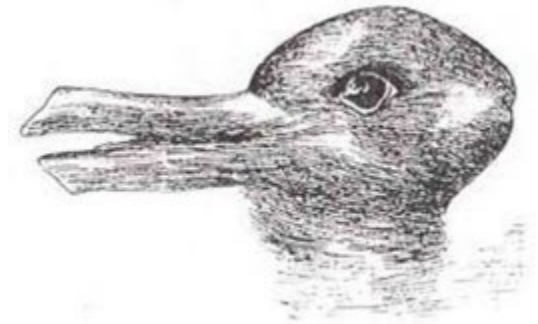


PERCEPTION: WE, LITERALLY, DO NOT SEE THE SAME I

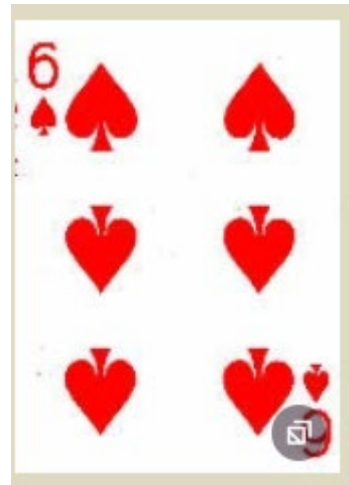
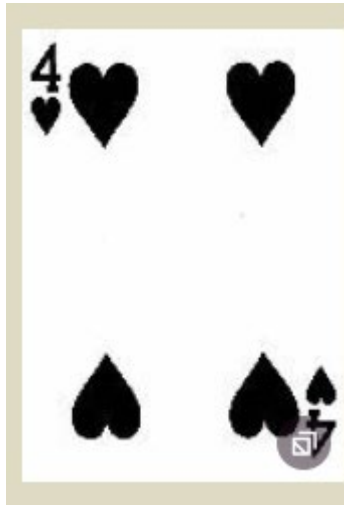
Data / the world does not imply just one interpretation, says Kuhn

- We don't see the world itself – just a processed version
- “In the metaphorical no less than in the literal use of 'seeing', interpretation begins where perception ends....what perception leaves for interpretation to **complete depends drastically on the nature and amount of prior experience and training**” (Kuhn p. 198, my emphasis).

≠ positivism

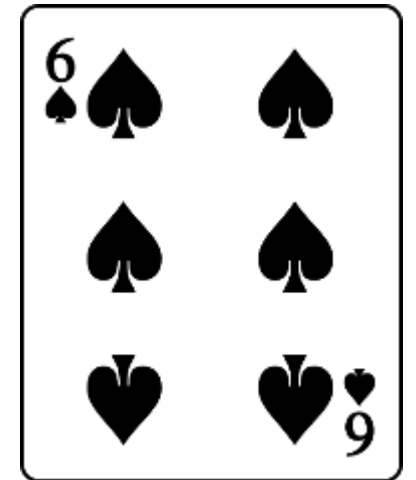
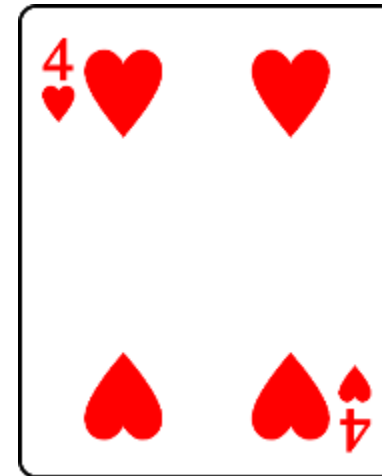
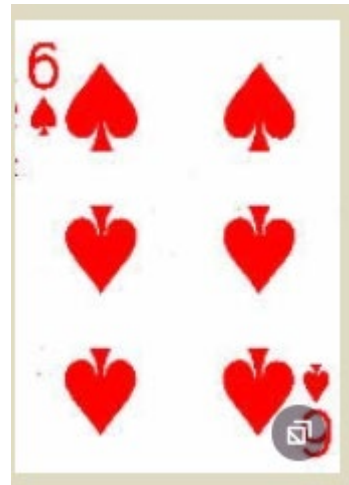
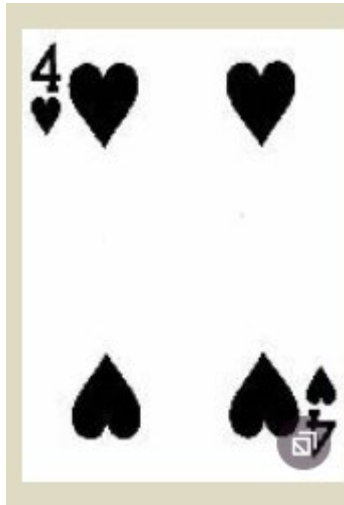


WHAT DO YOU SEE?



https://scientificmethod.fandom.com/wiki/Perception_is_Selective

WHAT DO YOU SEE?



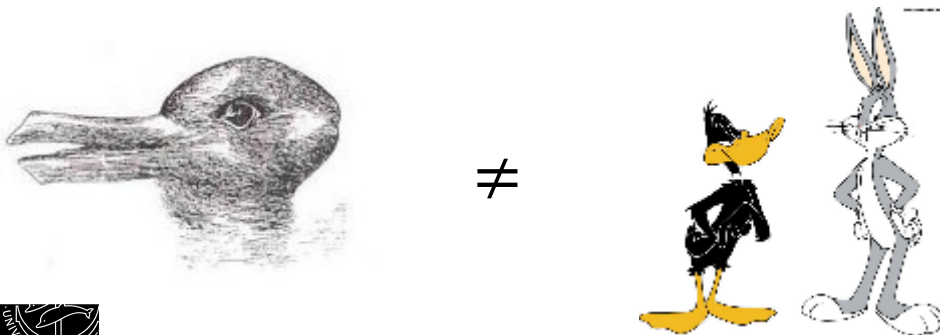
https://scientificmethod.fandom.com/wiki/Perception_is_Selective

Wikipedia

PERCEPTION: WE, LITERALLY, DO NOT SEE THE SAME II

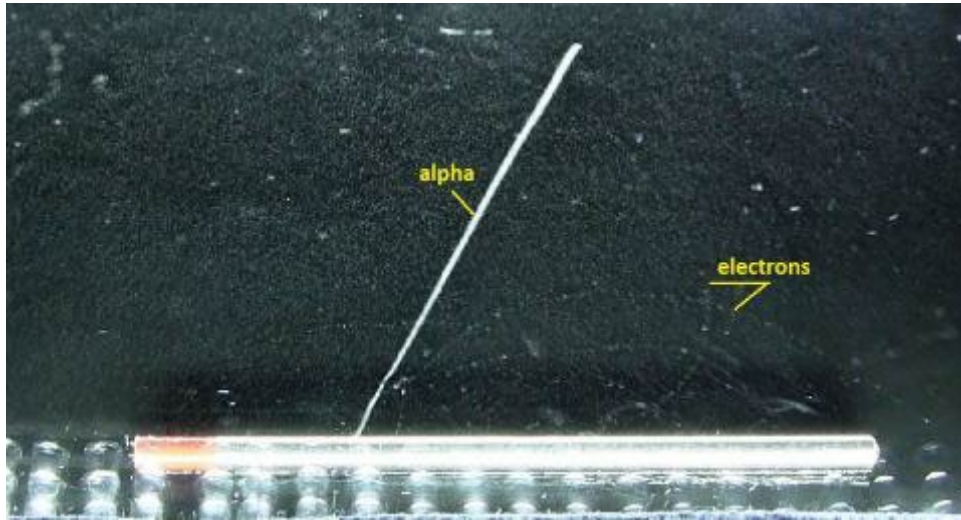
"If two people stand at the same place and gaze in the same direction, we must...conclude that they receive closely similar ...**But people do not see stimuli...**Instead they have sensations, and we are under no compulsion to suppose that the sensations of two viewers are the same...**much neural processing takes place between the receipt of a stimulus and the awareness of a sensation...**the route from stimulus to sensation is **in part conditioned by education.**" (Kuhn p. 192-193, my emphasis)

Although, one can't make up any kind of interpretation: Ducks and rabbits are different...



- Outside world creates stimuli
- Humans don't 'observe' stimuli – but experience already processed version of stimuli
- Different humans have different experiences
- Therefore: No neutral observation of the world

P. 196: "WE DO NOT *SEE* ELECTRONS, BUT RATHER THEIR TRACKS OR ELSE BUBBLES OF VAPOR IN A CLOUD CHAMBER".



https://upload.wikimedia.org/wikipedia/commons/a/ac/Alpha_particle_and_electrons_from_a_thorium_rod_in_a_cloud_chamber.jpg

- We don't directly 'see' motivation, efficient market hypothesis, productivity, firm performance etc.
- Paradigm --> perception

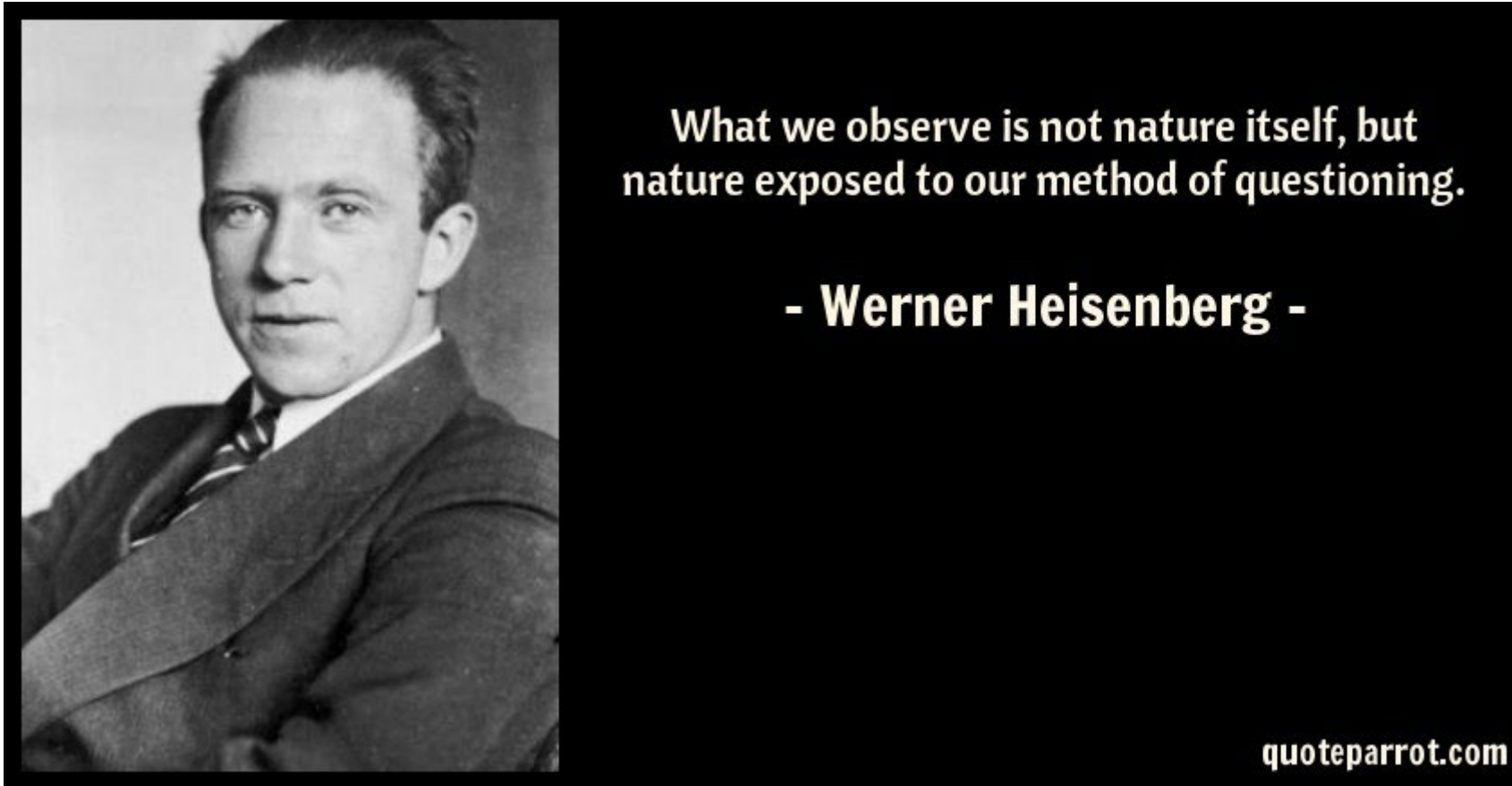
"In the metaphorical no less than in the literal use of 'seeing', interpretation begins where perception ends. The two processes are not the same, and what perception leaves for interpretation to complete depends drastically on the nature and amount of prior experience and training." (Kuhn p. 198)

"What we see depends in part on what we believe" (PoS p. 88)

"Facts are never just facts, facts are embedded within a paradigm" (PoS p. 84)



WE ONLY HAVE INDIRECT ACCESS TO REALITY / NATURE (HEISENBERG: QUANTUM PHYSICIST) ≠ positivism



KUHN'S PARADIGM SHIFTS ILLUSTRATED



<http://www.youtube.com/watch?v=3cp6pEzx3uw>

SOCIAL SCIENCE EXAMPLES: WE DON'T DIRECTLY SEE THE WORLD AS IT IS – BUT THROUGH A "PARADIGMATIC" LENS



If this is right – how can we be certain our theories actually represent the world / are true?

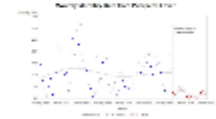
SCIENTIFIC NORMS → SHAPE WHAT EVIDENCE IS CONSIDERED

OVERALL: PARADIGMS SHAPE WHAT COUNTS AS GOOD EVIDENCE (PERCEPTION AND NORMS)

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- Sociology: Norms shape what is considered evidence

In other words: How researchers are stubborn and are not always convinced by evidence presented to others.



Cf. Ritchie 2020 (optimistic take on norms)

- Communalism: share results
- Universalism: evaluate claims, not people
- **Disinterestedness**: truth over self-interest
- **Organized skepticism**: criticism is a duty

→ Two competing scientific perspectives should find common ground

Kuhn (pessimistic/realistic (?) take on norms)

- Scientific communities reinforce existing beliefs (groupthink)
- Evidence can be ignored
- Science changes only slowly

→ Two competing scientific perspectives might not find common ground

MORE CONCRETELY: HOW SCIENTIFIC PRACTICES CAN REINFORCE EXISTING CONSENSUS

- **Description of science review process (cf. Ritchie 2020 p. 18-20)**
 - Researcher produces research
 - Submits to scientific journal
 - Editors sends out to 2-3 anonymous reviewers
 - Accept or reject (most submissions rejected)
 - Paper published in journal

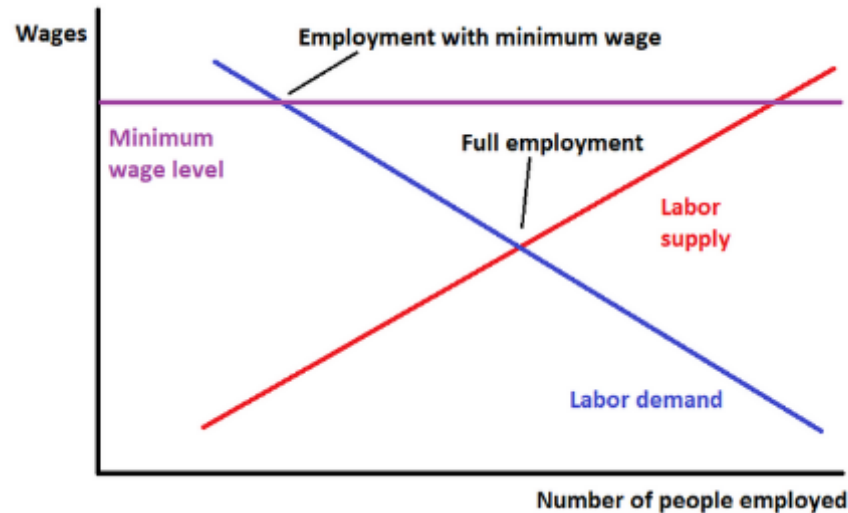
Risks enforcing existing consensus

- Journal/editor assess article based on existing standards / methods → conformity
- Past expertise shapes future review / editorial decisions → path dependency
- The more conforming science is more likely to become accepted?

Such a review / publication process likely to uphold status quo

EXAMPLE OF (TOO?) SLOW SCIENCE CHANGE: IMPACT OF MINIMUM WAGE

1994 landmark (Nobel prize winning) study claims: Increasing minimum wage did not reduce employment



An example of a fundamental change in assumptions about the world

<https://noahpinion.substack.com/p/why-15-minimum-wage-is-pretty-safe>

STUBBORN REACTION FROM RESEARCH COMMUNITY OF COURSE MINIMUM WAGE WOULD NEGATIVELY IMPACT EMPLOYMENT...RIGHT?

“The inverse relationship between quantity demanded and price is the core proposition in economic science, which embodies the presupposition that human choice behavior is sufficiently rational to allow predictions to be made. **Just as no physicist would claim that “water runs uphill,” no self-respecting economist would claim that increases in the minimum wage increase employment.** Such a claim, if seriously advanced, becomes equivalent to a denial that there is even minimal scientific content in economics, and that, in consequence, economists can do nothing but write as advocates for ideological interests. **Fortunately, only a handful of economists are willing to throw over the teaching of two centuries; we have not yet become a bevy of camp-following whores.”**

~James M. Buchanan, 1986 Nobel laureate in economics, writing in the Wall Street Journal on April 25, 1996

SLOWLY CHANGING BELIEF ON EFFECT OF MINIMUM WAGE

Changing opinion among economists:

Do minimum wages substantially lower employment among low-wage workers?

- 1978 AEA Member Survey: **90%** agreed
- 1992 AEA Member Survey: **72%** agreed
- 2000 AEA Member Survey: **46%** agreed
- 2013 IGM Panel (\$9/hr): **34%** agreed
- 2015 IGM Panel (\$15/hr): **26%** agreed

Analysis of petition signers (O'Neill 2014):

Labor economists, recent PhDs *more* likely to support raising minimum wages

<https://noahpinion.substack.com/p/why-15-minimum-wage-is-pretty-safe>

CARD'S (2021 NOBEL PRIZE WINNER) EXPERIENCE

HOW NORMS SHAPE THE (I)RATIONALITY OF DISCUSSION?

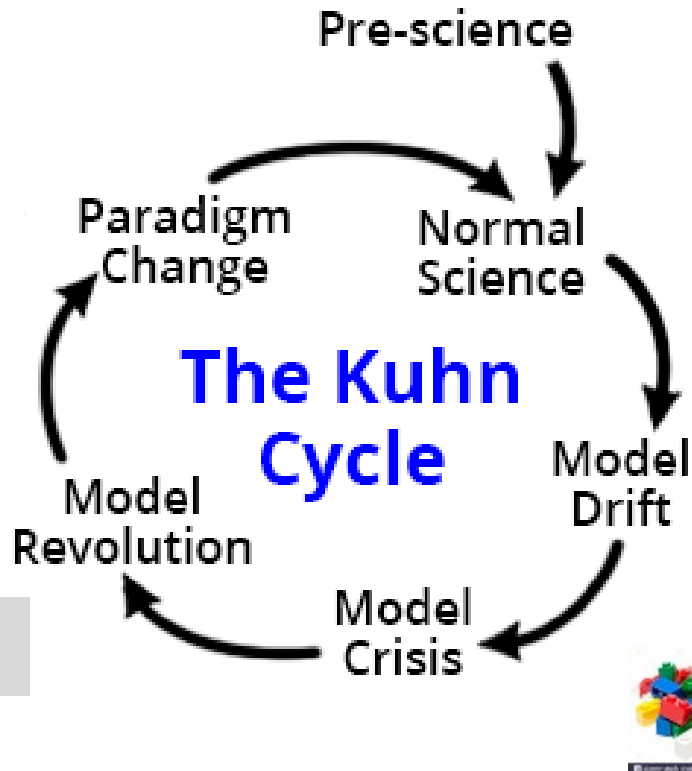
“[E]conomists who objected to our work were upset by the thought that we were giving free rein to people who wanted to set wages everywhere at any possible level...I've subsequently stayed away from the minimum wage literature for a number of reasons. First, it cost me a lot of friends. People that I had known for many years, for instance, some of the ones I met at my first job at the University of Chicago, became very angry or disappointed. They thought that in publishing our work we were being traitors to the cause of economics as a whole.”

<https://noahpinion.substack.com/p/why-15-minimum-wage-is-pretty-safe>

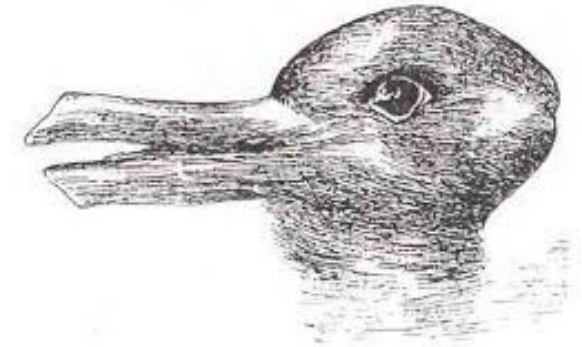
- Changes in fundamental (paradigmatic) assumptions and beliefs are difficult, stubborn and slow
- Science not a fully rational process / institution
- Example illustrates Kuhn quote: **“A revolution is for me a special sort of change involving a certain sort of reconstruction of group commitments”** (Kuhn, p. 181, see also p. 82-83 in PoS)

KUHN'S TAKE ON SCIENTIFIC DEVELOPMENT

(REFLECT, DISCUSS 2 MIN – QUESTIONS?)



Typical (normal) science



Revolutions are –
very – rare!

"A revolution is for me a special sort of change involving a certain sort of reconstruction of group commitments" (Kuhn, p. 181, see also p. 82-83 in PoS)

<http://www.thwink.org/sustain/glossary/KuhnCycle.htm>

SUMMING UP: A SCIENTIFIC PARADIGM

Paradigms showcase a research agenda

- What is important
- Which skills a researcher should have
- Which data are collected
- What is being taught
- What research is being accepted
-

The concept explains why research changes slowly – difficult to change these established/consensus based practices

Theories can more easily change -
paradigms are embedded practices that
are difficult to change

SO, HOW DO PARADIGMS CHANGE AND IMPROVE AFTER ALL, ACCORDING TO KUHN?

If science can't improve and two paradigms just are incomparably different →
pure relativism, anything goes?

Kuhn's answer: About (lack of) truth, funerals and survival of the fittest

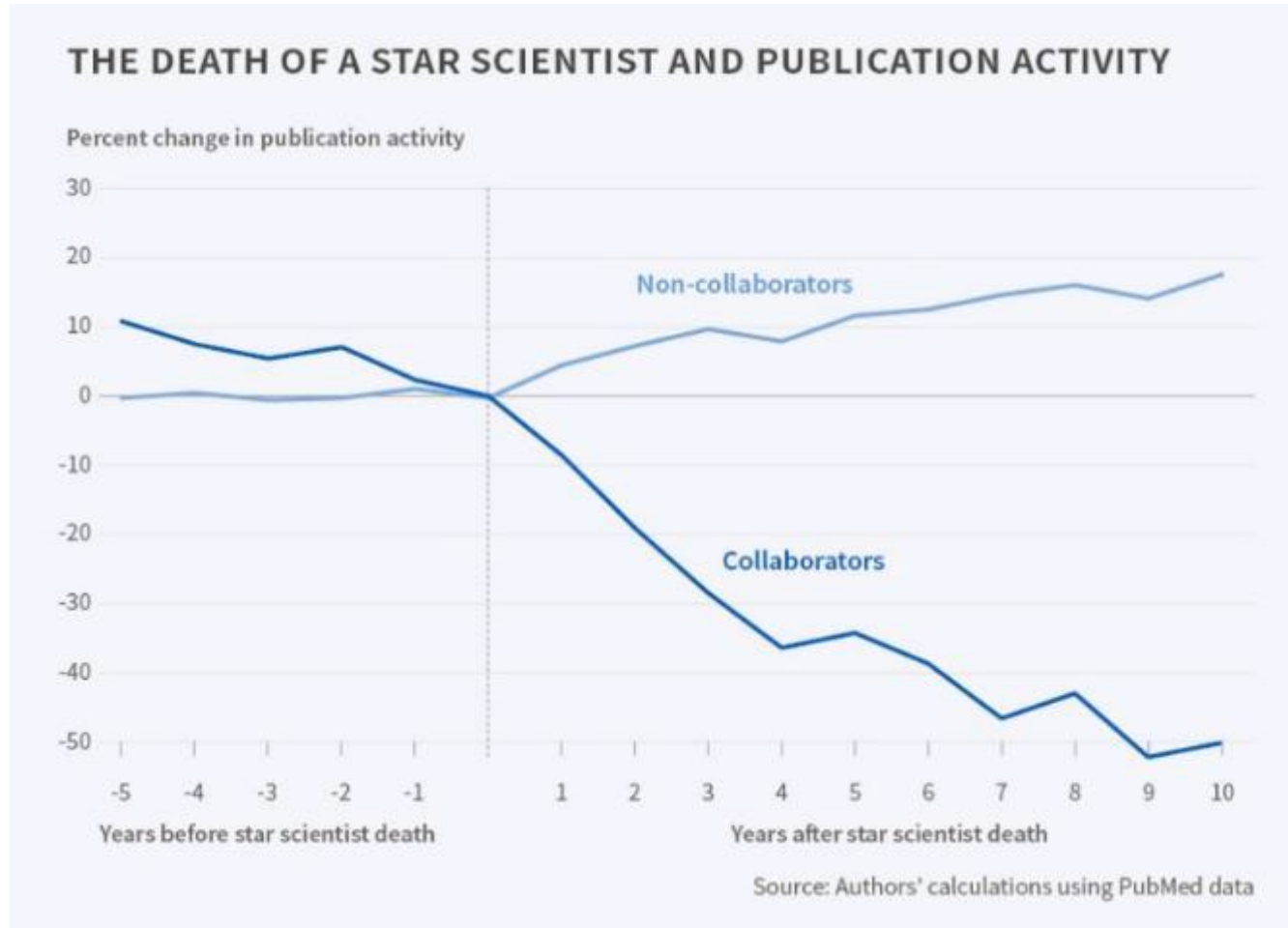
A new scientific truth does not triumph by convincing its opponents and making them see the light, but rather because its opponents eventually die and a new generation grows up that is familiar with it ...

An important scientific innovation rarely makes its way by gradually winning over and converting its opponents: it rarely happens that Saul becomes Paul. What does happen is that its opponents gradually die out, and that the growing generation is familiarized with the ideas from the beginning: another instance of the fact that the future lies with the youth.

— Max Planck, Scientific autobiography, 1950, p. 33, 97

https://en.wikipedia.org/wiki/Planck%27s_principle

SCIENCE, ONE FUNERAL AT A TIME? (SHIFTING NORMS)



What research design
is this?

What to pay attention
to?

NEW PARADIGMS ARE NOT MORE TRUE I

Kuhn p. 206: There is, I think, **no theory-independent** way to reconstruct phrases like ‘really there’; **the notion of a match between the ontology of a theory and its “real” counterpart in nature now seems to me illusive in principle**” (my emphasis. A clear critique of positivism)

POS, p. 84: “...Kuhn questioned whether the concept of objective truth actually makes sense at all. The idea that there is a fixed set of facts about the world, independent of any particular paradigm, was of dubious coherence, he believed. Kuhn suggested a radical alternative: **That facts about the world are paradigm-relative, and thus change when paradigms change. If this suggestion is correct, then it makes no sense to ask whether a given theory corresponds to the facts ‘as they really are’, nor therefore to ask whether it is objectively true.**” (my emphasis)

NEW PARADIGMS ARE NOT MORE TRUE II

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Kuhn p. 204: “...as argument piles on argument and as challenge after challenge is successfully met, only blind stubbornness can at the end account for continued resistance”.

SCIENTISTS DO (ACTUALLY) SOMETIMES CHANGE THEIR MINDS



Erik Brynjolfsson ✓
@erikbryn



"In science, it often happens that scientists say: 'You know that's a really good argument; my position is mistaken,'

And then they would actually change their minds and you never hear that old view from them again.

They really do it.

It doesn't happen as often as it should, because scientists are human and change is sometimes painful. But it happens every day.

I cannot recall the last time something like that happened in politics or religion."

-Carl Sagan

COMPARING DIFFERENT PARADIGMS → SURVIVAL OF THE FITTEST

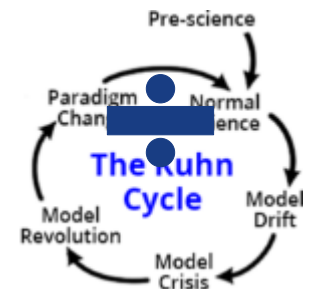
- Not only about funerals – newer paradigms can be better / not simple relativism
- Yet not better representation of reality, cf. Kuhn 1970: p. 206

→ survival of the fittest, across paradigms

- Criteria for comparison (Kuhn 1970: p. 206)
 - Accuracy of prediction
 - Number of different problems solved
 - Simplicity, scope, compatibility with other disciplines



”...scientific development is, like biological, a unidirectional and irreversible process. Later scientific theories are better than earlier ones for solving puzzles in the often quite different environments to which they are applied” (Kuhn 1970: p. 206)



WHICH QUOTE IS NOT BY KUHN?

(VEVOX. APP 143-136-950, 2 MIN)

- 1) Later scientific theories are better than earlier ones for solving puzzles in the often quite different environments to which they are applied. That is not a relativist's position, and it displays the sense in which I am a convinced believer in scientific progress.
- 2) 'Normal science' means research firmly based upon one or more past scientific achievements, achievements that some particular scientific community acknowledges for a time as supplying the foundation for its further practice.
- 3) A new paradigm in the end wins over the former, established paradigm, by developing new theories and practices that manage to obtain a clearer and better understanding of reality as it is.

QUESTION FROM 2018 RE-EXAM

Question 3 (15%)

Please explain why Kuhn would NOT agree with the following statement: “A new paradigm in the end wins over the former established paradigm, by developing new theories and practices that manage to obtain a clearer and better understanding of reality as it is.”

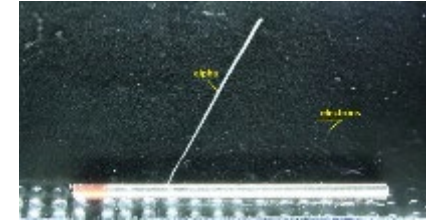
Please be specific about what part of the statement Kuhn would disagree with, and explain why.

ANSWER TO Q3 FROM 2018 RE-EXAM

A FEW BULLET-POINTS TO ILLUSTRATE WHAT A VERY GOOD ANSWER CAN LOOK LIKE

- To see the world as it is = positivism (Guba 1990, see lesson 9)
- "There is, I think, no theory-independent way to reconstruct phrases like "really there"(Kuhn 1970, p. 206)
- We see reality through our paradigmatic (values, techniques, etc. that members of our scientific community share) lenses
 - Interpretation begin where perception ends (Kuhn p. 198)
- Newer, better theories not approximately closer to reality (Kuhn p. 206)
 - As post-positivism would claim (Guba 1990, lesson 8)
- Different paradigms are incommensurable
 - Difficult to communicate across: "Since the vocabularies in which they discuss such situations consist, however, predominantly of the same terms, they must be attaching some of those terms to nature differently, and their communication is inevitably only partial"(Kuhn 1970, p. 198)
- But, in an evolutionary perspective: Better at solving problems, more accurate predictions, solves more and more important problems (Kuhn p. 206)
 - One paradigm might only be able to explain lab-experiment, while another can explain phenomena in the field
 - Thus, a paradigm can win over another – but not an ontological argument
 - "as argument piles on" (Kuhn p. 204)
- Therefore, since different conceptual frameworks and ways of seeing the world, it is not the world itself that is being compared, but paradigmatic approaches: Some are better than others, thus not simple relativism/subjectivism.

KUHN: SUMMING UP



Kuhn: A history of science perspective

- Keywords: Paradigms, normal science, perception, norms, revolutions, scientific community
- How have the sciences developed, historically – a psychological-sociological perspective
- We – literally – don't see the same: Depends on training/ education/ expectations.
- Not achieving 'truth': Evolutionary perspective $\neq$ linear, brick-by-brick development

Kuhn (my) take-away:

- Simple positivism doesn't explain science change
- Truth is a tricky concept
- Yet, Kuhn a bit too close to simple "relativism" & very different researchers can still communicate/disagree/learn from each other...

THE (POSITIVISM) POSITION THAT KUHN & POPPER ARGUE AGAINST



Positivism: Classic textbook conception of science pre-Kuhn (1970)

- "positivists believed that disputes between rival scientific theories could be solved in a perfectly objective way, comparing the theories directly to the observations..." (ToS 5, p.80)
- "the world can be observed as it is / there are neutral facts" (ToS p. 80)
- "Theory corresponds to / mirrors world" (ToS p. 70)
- "We can tell if a theory is true / false"
- "Theory development is a rational process" (ToS p. 78-79)

POPPER AND KUHN - ON SCIENCE, OBJECTIVITY AND SETTLING DISAGREEMENTS

PHILOSOPHY OF SCIENCE: HOW DO THEORIES DEVELOP AND IMPROVE OVER TIME?

Not about one individual theory – but the **philosophy of science** (in general)
Insights on research design & deduction and induction can give impression of

- Science directly building on prior work
- Science continuously improving
- We can observe the world as it is
- We can test theories & verify truth

→ Positivism

Science as a very rational, cumulative process



THE (POSITIVISM) POSITION THAT KUHN & POPPER ARGUE AGAINST



Positivism: Classic 'textbook' conception of science pre-Kuhn (1970)

- "...positivists believed that disputes between rival scientific theories **could be solved in a perfectly objective way**, comparing the theories directly to the observations..." (PoS 5, p.80)
 - The **world can be observed as it is** / there are neutral facts (PoS p. 80)
 - **Theory corresponds to / mirrors world** (PoS p. 78) →
 - We can tell if a theory is true / false
 - Theory development as a **rational process** (PoS p. 78-79)

- **Shared perspective:** no theory-neutral observation of the world
- **Kuhn (descriptive / historical):** how science often changes in practice and survival of the fittest
- **Popper (normative / method):** how science should develop → falsification

POPPER 1962

Popper agenda

- Observing the world as it is & and problem with induction
- Asymmetry
- Verification vs. falsification of theories
- No absolute truth, only tentative (-> post-positivism, more in lesson 9)

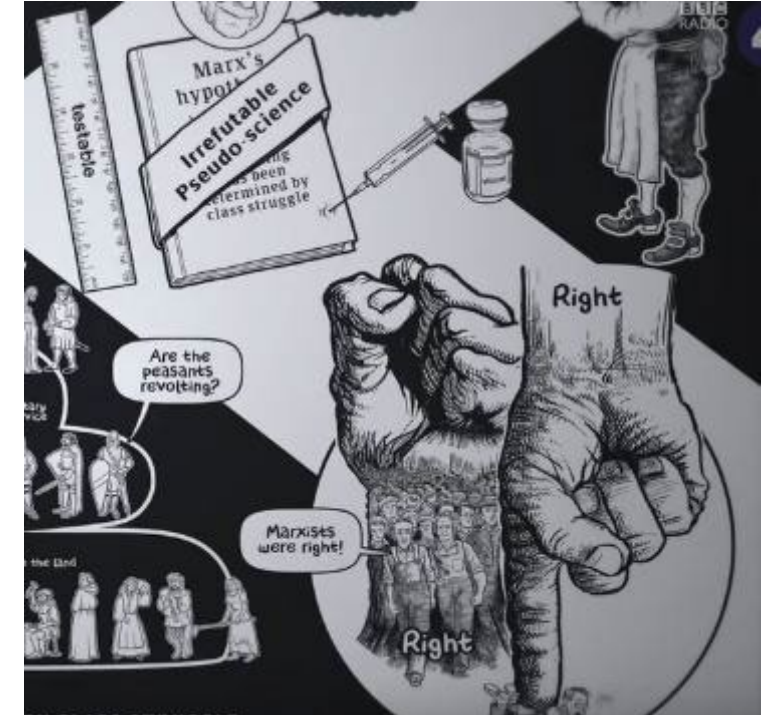
Overall question: How do we investigate theories, and can theories be true?

E.g.: can induction or deduction lead to truth?

VIDEO: FALSIFICATION



VIDEO: FALSIFICATION II



POPPER – TO BE CONTINUED IN LESSON 8